IIoT Network Analysis: Age of Information and Reliability Trade-offs

Summary Report

Judith Barrios — ITAI3377

L07_Report_Judith_Barrios_ITAI3377

Overview

This report summarizes the main findings from the L07 notebook assignment, which focused on Age of Information and reliability tradeoffs in Industrial Internet of Things networks using a simulated dataset of 10,000 records and the theoretical framework from Farag, Ali, and Stefanovic (2023). Out of those 10,000 records, 1,397 had infinite AoI values meaning the packets never got through at all, so the machine learning portion worked with the remaining 8,603 rows that had real finite values to train and test on. Sisinni et al. (2018) give a useful overview of why this kind of analysis matters for real industrial deployments and what the main communication challenges look like once these networks are actually running in a factory or plant environment.

Conceptual Background

Age of Information is a way of measuring how old the data sitting at the receiving end of a network actually is at any given moment, and what makes it different from older metrics like latency or throughput is that those only tell you about one packet at a time moving through the system, whereas AoI tells you whether the controller has information that actually reflects the current state of the process rather than a state from several seconds ago. Sun et al. (2017) pointed out that trying to minimize delay or maximize how much data gets through does not automatically lead to the freshest possible information at the destination, which is a big part of why AoI developed into its own research area and why a system can look perfectly healthy by traditional measures and still leave a controller working with stale data.

The Farag et al. (2023) paper is built around the idea that most real industrial wireless networks carry two very different types of traffic at the same time, and most earlier research only looked at one of them without considering how they affect each other. AoI-oriented traffic is regular monitoring data from sensors sending readings at fixed intervals, where a failed message just gets replaced by the next fresh reading rather than retransmitted since a newer value is already ready. Deadline-oriented traffic covers emergency alerts that must arrive within a hard time window or they become useless, so those devices keep retrying until something gets through or the window closes, and Packet Loss Probability tracks how often they miss that window entirely. The wireless channel model in the paper is built on the capture effect from Zorzi and Rao (1994), where a message can still be decoded under interference if the signal is strong enough to clear a certain threshold, and Farag et al. (2019) found similar tradeoff patterns in other mixed-traffic industrial sensor network setups.

Data Exploration Findings

One of the more interesting things from looking at the visualizations was that sending messages more often does not simply push AoI down in a straight line the way you might expect it to. At very low transmission rates the information age stays high because updates come in only rarely, but once the rate climbs high enough that many devices are sending at the same time, the AoI starts rising again because so many competing signals are getting in each other's way that fewer messages actually get decoded successfully. This matches the paper's formula where AoI equals one divided by the successful delivery probability, so anything driving that probability down will raise the information age even if devices are working harder than ever to send, which is exactly the non-obvious relationship Maatouk et al. (2020) argue makes AoI worth treating as its own metric rather than just a side effect of delay.

The correlation heatmap showed that channel quality had the strongest relationship with information age out of any variable in the dataset, which lines up directly with the capture effect from Zorzi and Rao (1994) that the paper's model is built on, since better channel conditions mean the signal-to-interference ratio is more likely to clear the threshold needed for a successful decode and more messages get through as a result. The visualizations also made it clear that having more devices competing for the same medium tends to make things considerably worse, and this effect gets especially severe when the wireless connection is already weak, with the worst AoI values in the entire dataset almost always appearing when both of those conditions were true at the same time.

Data Visualizations

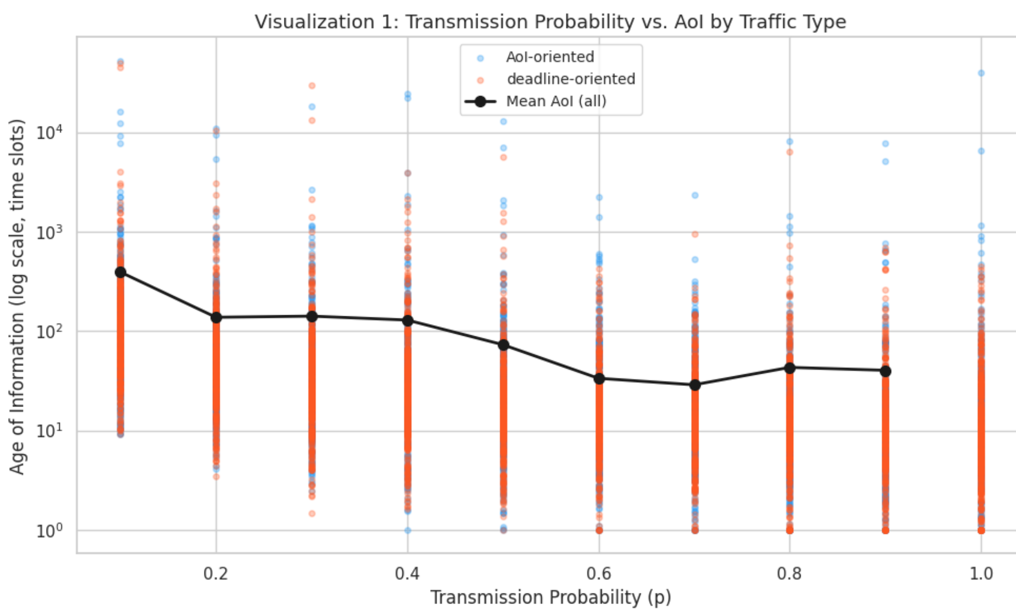

Figure 1: Transmission Probability vs AoI by Traffic Type (log scale)

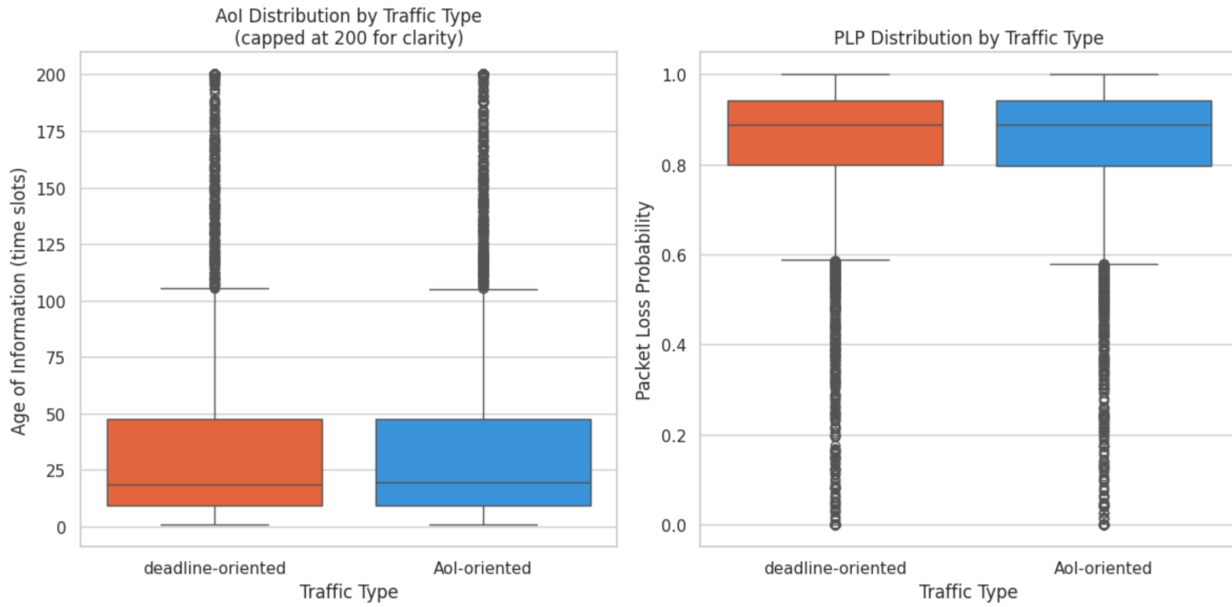

Figure 2: AoI and PLP Distribution by Traffic Type

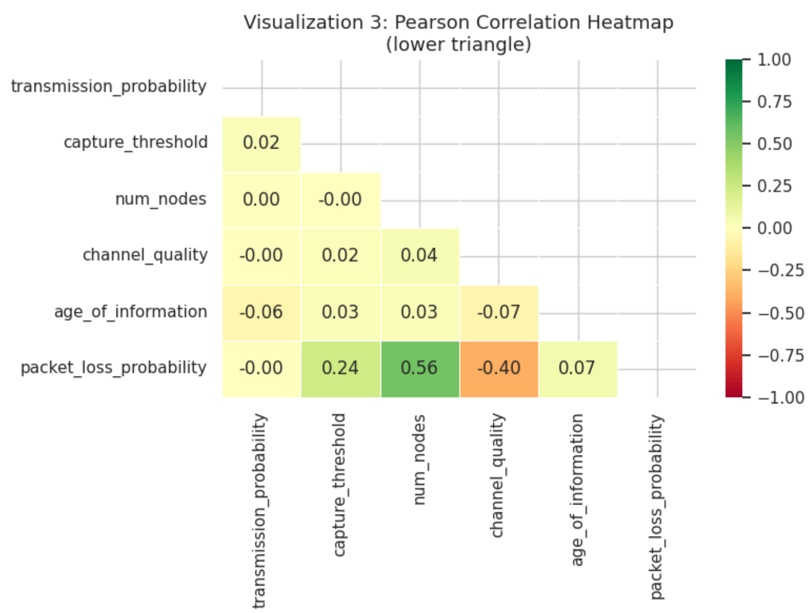

Figure 3: Pearson Correlation Heatmap

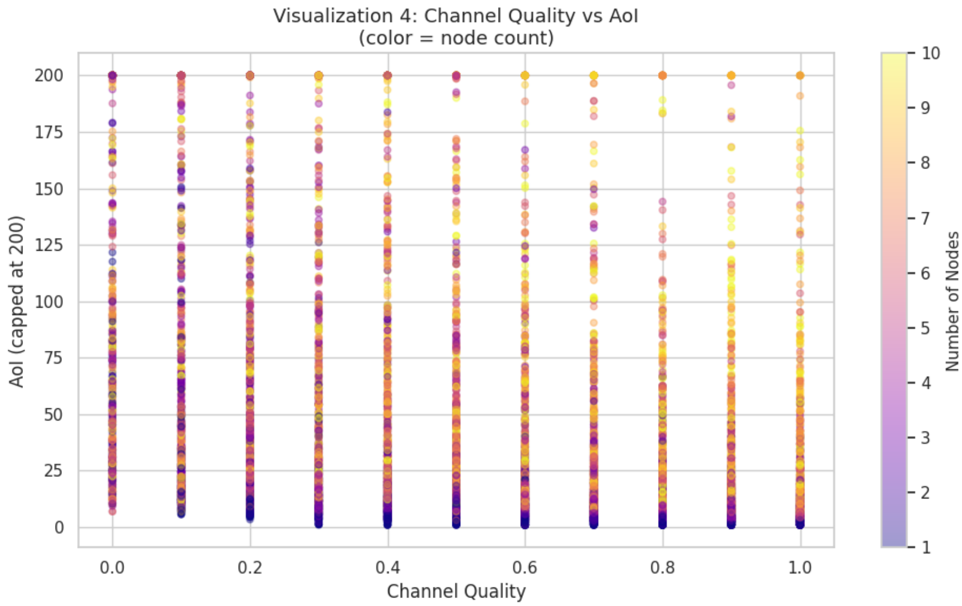

Figure 4: Channel Quality vs AoI colored by Node Count

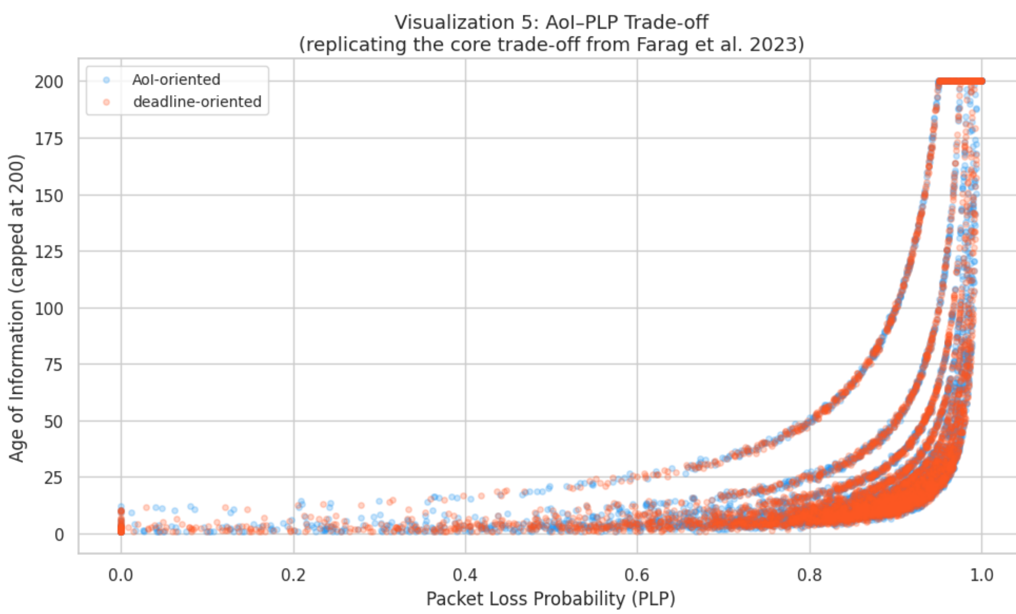

Figure 5: AoI vs PLP Trade-off by Traffic Type

Machine Learning Results

The Random Forest came back with an R squared of -0.37 and an RMSE of 1,736 time slots, which is not a good result and is worth being upfront about rather than glossing over. A negative R squared means the model actually did worse than just predicting the average AoI for every single row, and the most likely explanation is the extreme spread in AoI values across the dataset where some configurations sit near 1 time slot while others reach into the hundreds, combined with the 1,397 infinite rows that had to be dropped before training and removed a meaningful chunk of information. Applying a log transformation to the AoI values before fitting the model would probably have helped considerably, but even with the poor overall accuracy the three hypothetical predictions still made intuitive sense, with Config A under ideal conditions returning 2.23 time slots, Config B under worst-case conditions returning 136 time slots, and Config C under moderate conditions returning 22.73 time slots.

The feature importance output was honestly more informative than the accuracy score for understanding what actually drives AoI in this kind of network. Capture threshold ranked first at 0.448 since it directly controls whether the receiver can decode a message under interference at all, number of nodes came second at 0.231, channel quality third at 0.171, transmission probability at 0.095, and traffic type last at 0.055, all of which is consistent with the theory from Farag et al. (2023). The deep learning model did slightly better on AoI prediction at an R squared of -0.20, but where it genuinely stood out was PLP prediction where it achieved an R squared of 0.91, which is a strong result that makes sense since PLP is bounded between 0 and 1 and has far less extreme variance than AoI, making it a much more learnable target.

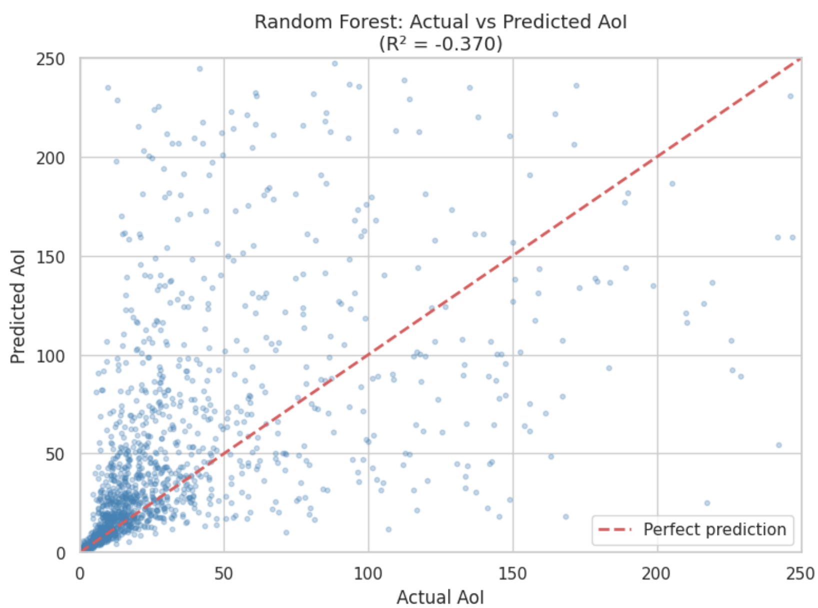

Figure 6: Random Forest - Actual vs Predicted AoI

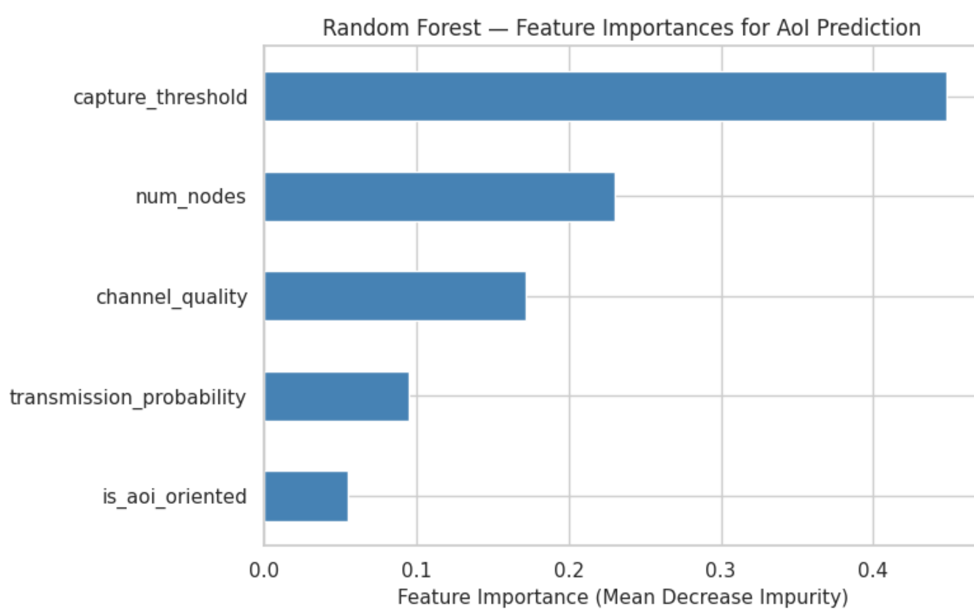

Figure 7: Random Forest - Feature Importances

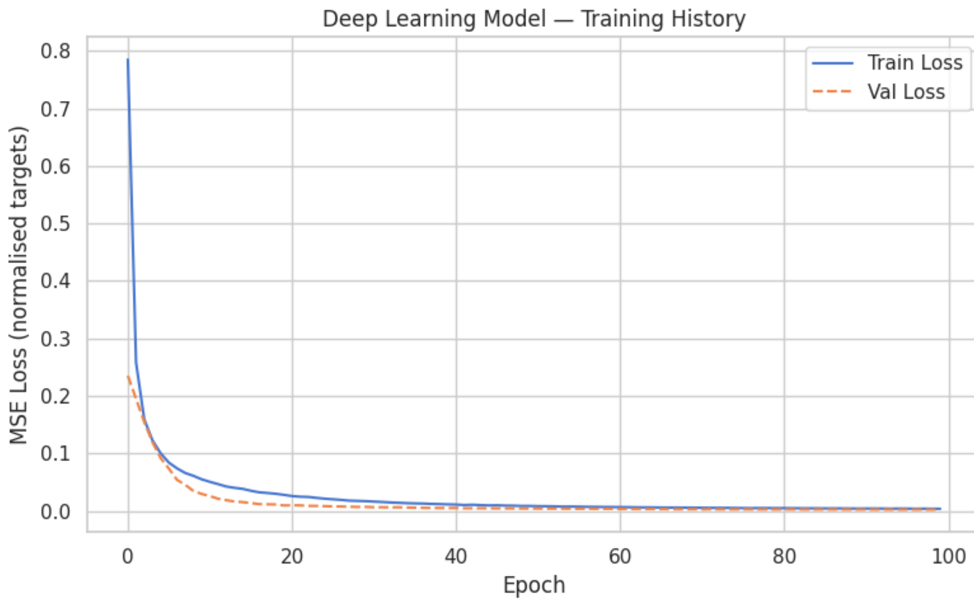

Figure 8: Deep Learning Model - Training History

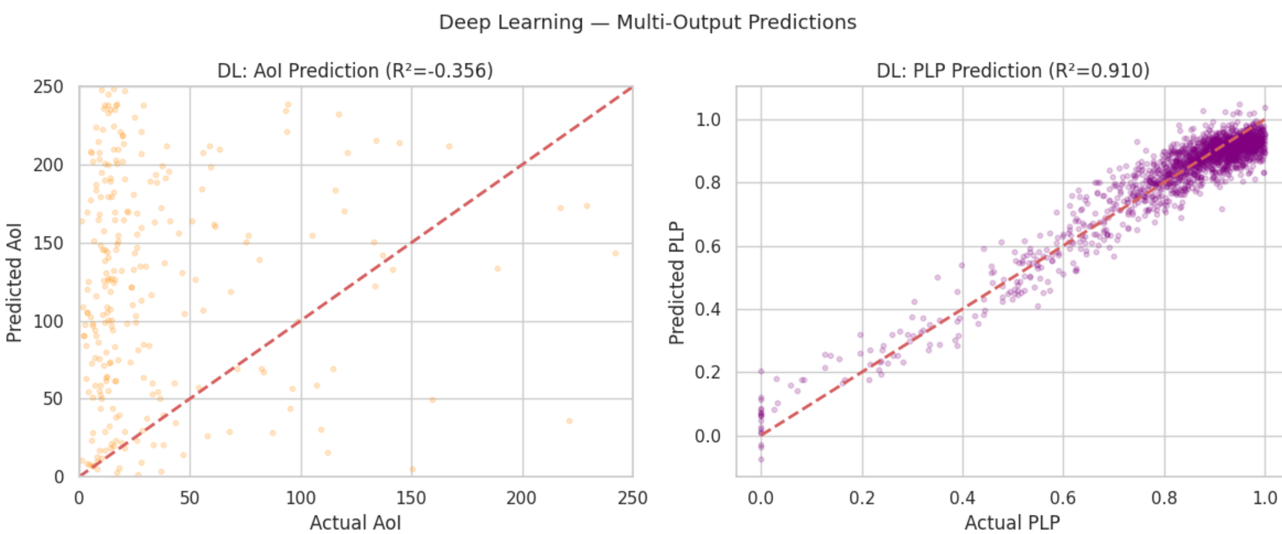

Figure 9: Deep Learning - Actual vs Predicted AoI and PLP

Key Insights and Strategies

Transmission probability turned out to be the main practical lever available to a network operator once a network is deployed, since channel quality and node count are mostly determined by the physical environment and layout and cannot easily be changed after the fact. Capture threshold had the highest feature importance in the model at 0.448, which suggests that investing in receiver hardware that can tolerate more interference would be one of the most effective improvements possible, and Figure 3 of Farag et al. (2023) shows that a more lenient threshold can cut the information age by more than half for the exact same transmission behavior without changing anything about how devices operate on the network.

Two strategies were proposed in the notebook for balancing freshness and reliability in practice. The first is to have monitoring devices adjust how often they send based on whether a deadline alert is currently in the queue, pulling back when an alarm needs a cleaner channel and returning to their normal rate once delivery is confirmed, so the network can shift along the tradeoff curve depending on what is most urgent at any given moment rather than being locked at one fixed point. The second is to upgrade the receiver hardware using interference cancellation technology to lower the capture threshold, which improves both AoI and PLP at the same time without requiring any change to device behavior. Power substation monitoring and factory robot safety systems were both discussed as real applications where getting this balance right has direct safety and cost consequences.

References

Farag, H., Ali, S. M., & Stefanovic, C. (2023). On the analysis of AoI-reliability tradeoff in heterogeneous IIoT networks. arXiv preprint arXiv:2311.13336.

Farag, H., Sisinni, E., Gidlund, M., & Osterberg, P. (2019). Priority-aware wireless fieldbus protocol for mixed-criticality industrial wireless sensor networks. IEEE Sensors Journal, 19(7), 2767-2780.

Maatouk, A., Kriouile, S., Assaad, M., & Ephremides, A. (2020). The age of incorrect information: A new performance metric for status updates. IEEE/ACM Transactions on Networking, 28(5), 2215-2228.

Sisinni, E., Saifullah, A., Han, S., Jennehag, U., & Gidlund, M. (2018). Industrial Internet of Things: Challenges, opportunities, and directions. IEEE Transactions on Industrial Informatics, 14(11), 4724-4734.

Sun, Y., Uysal-Biyikoglu, E., Yates, R. D., Koksal, C. E., & Shroff, N. B. (2017). Update or wait: How to keep your data fresh. IEEE Transactions on Information Theory, 63(11), 7492-7508.

Zorzi, M., & Rao, R. (1994). Capture and retransmission control in mobile radio. IEEE Journal on Selected Areas in Communications, 12(8), 1289-1298.